

Deep Research Agents Overview and Implementations

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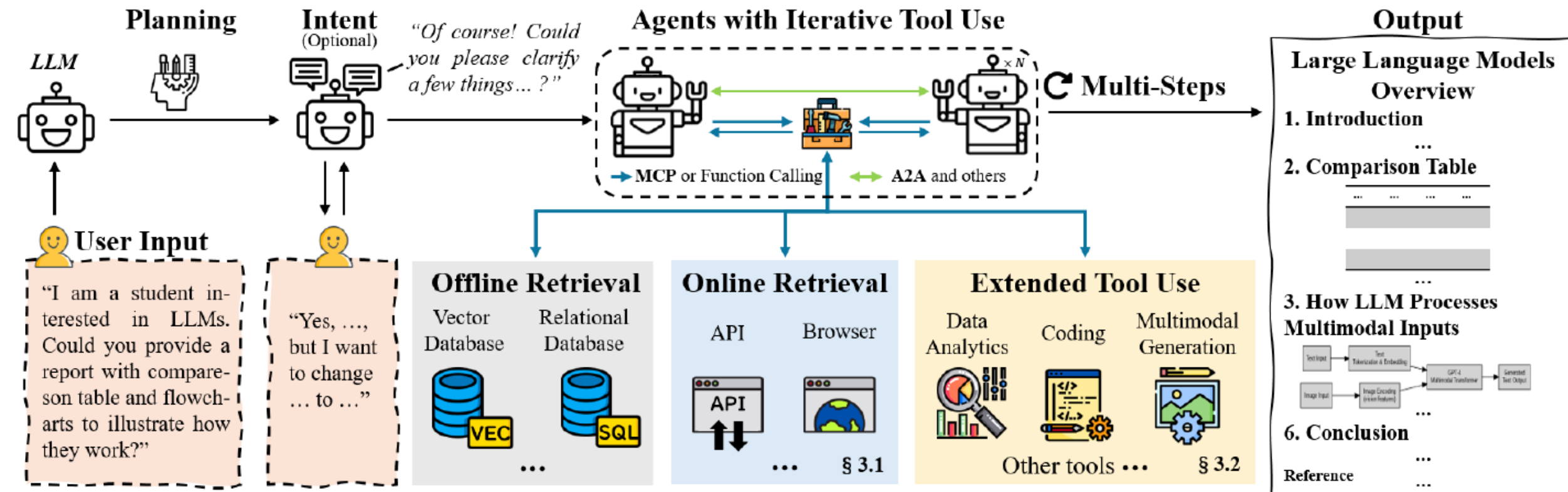
Deep Research (DR) Agents



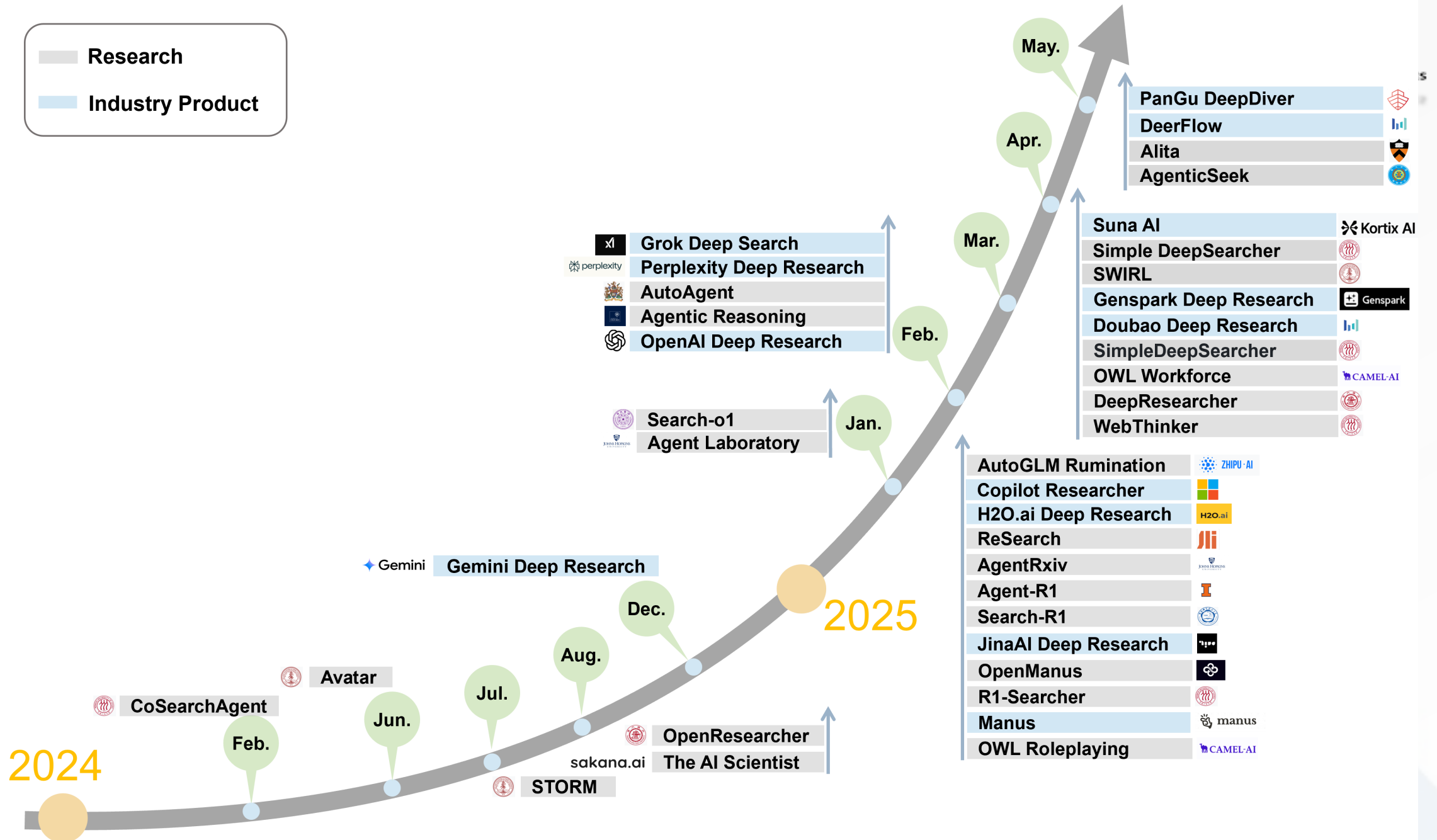
AI agents powered by **LLMs**, integrating dynamic **reasoning**, adaptive **planning**, **multi-iteration** external data **retrieval** and **tool use**, and comprehensive analytical **report generation** for informational research tasks.

- Traditional RAG: Enhance factual accuracy but **lack** sustained **reasoning** capabilities
- Tool Use (TU) systems: Heavily depend on **pre-defined** workflow

Deep Research Agents: Overview



An overview of a DR agent

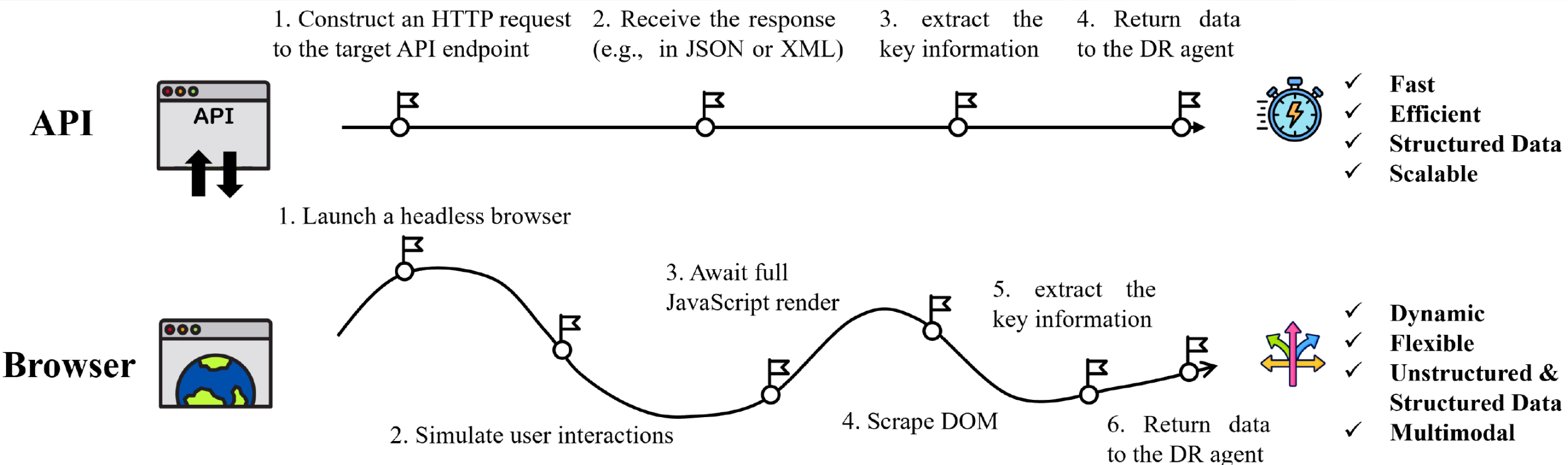


Deep Research Agents: Overview

- Search engine
- Tool Use
- Workflow
- Tuning
- Non-parametric continual learning

Search Engine: API vs. Browser

- API, interact with **structured** data sources (Google Search, arXiv, Wikipedia, X's native interface, Semantic Scholar, SerpApi (slack-based), PubMed, ClinicalTrials.gov, Bing Search, Jina Reader)
- Browser, **human-like** interactions with web pages (sandboxed Chromium, BrowserGym)



Search Engine

- Benefit from the **efficiency** of API-based methods, with the **comprehensiveness** of browser-driven exploration.

Table 1: Comparison of DR Agents with Search Engine Details

■ = Primary focus, ■ = Secondary/minor focus, □ = Not present

DR Agent	Search API	Engine Browser	GAIA	HLE	Benchmark Other QA	Base Model	Release
Avatar [106]	■	□	□	□	Stark	Claude-3-Opus, GPT-4	Feb-2024
CoSearch-Agent [28]	■	□	□	□	□	GPT-3.5-turbo	Feb-2024
MMAC-Copilot [87]	■	□	■	□	□	GPT-3.5, GPT-4	Mar-2024
Storm [81]	■	□	□	□	FreshWiki	GPT-3.5-turbo	Jul-2024
OpenResearcher [118]	■	□	□	□	Privately Collected QA Data	DeepSeek-V2-Chat	Aug-2024
The AI Scientist [57]	■	□	□	□	MLE-Bench	GPT-4o, o1-mini, o1-preview	Aug-2024
Gemini DR [30]	■	■	□	■	GPQA	Gemini-2.0-Flash	Dec-2024
Agent Laboratory [79]	■	□	□	□	MLE-Bench	GPT-4o, o1-preview	Jan-2025
Search-o1 [52]	■	□	□	□	GPQA, NQ, TriviaQA	QwQ-32B-preview	Jan-2025
Agentic Reasoning [105]	■	□	□	□	GPQA	DeepSeek-R1, Qwen2.5	Feb-2025
AutoAgent [91]	□	■	■	□	□	Claude-Sonnet-3.5	Feb-2025
Grok DeepSearch [107]	■	■	□	□	GPQA	Grok3	Feb-2025
OpenAI DR [69]	□	■	■	■	■	GPT-o3	Feb-2025
Perplexity DR [72]	■	■	□	■	SimoleQA	Flexible	Feb-2025
AgentRxiv [78]	■	□	□	□	GPQA, MedQA	GPT-4o-mini	Mar-2025
Agent-R1 [71]	■	□	□	□	HotpotQA	Qwen2.5-1.5B-Inst	Mar-2025
AutoGLM Rumination[119]	□	■	□	□	GPQA	GLM-Z1-Air	Mar-2025
Copilot Researcher [63]	□	■	□	□	□	o3-mini	Mar-2025
H2O.ai DR [35]	■	■	■	□	□	h2ogpt-oasst1-512-12b	Mar-2025
Manus [60]	■	■	□	□	□	Claude3.5, GPT-4o	Mar-2025
Openmanus [54]	■	■	□	□	□	Claude3.5, GPT-4o	Mar-2025
OWL [11]	■	■	■	□	□	Deerpeek-R1, Gemini2.5-Pro, GPT-4o	Mar-2025
R1-Searcher [86]	■	□	□	□	2WikiMultiHopQA, HotpotQA	Llama3.1-8B-Inst, Qwen2.5-7B	Mar-2025
ReSearch [15]	■	□	□	□	2WikiMultiHopQA, HotpotQA	Qwen2.5-7B, Qwen2.5-7B-Inst	Mar-2025
Search-R1 [47]	■	□	□	□	2WikiMultiHopQA, HotpotQA, NQ, TriviaQA	Llama3.2-3B, Qwen2.5-3B/7B	Mar-2025
DeepResarcher [117]	□	■	□	□	HotpotQA, NQ, TriviaQA	Qwen2.5-7B-Inst	Apr-2025
Gensparks Super Agent [94]	■	■	■	□	□	Mixture of Agents*	Apr-2025
WebThinker [53]	■	□	■	■	GPQA, WebWalkerQA	QwQ-32B	Apr-2025
SWIRL [27]	□	■	□	□	HotQA, BeerQA	Gemma 2-27b	Apr-2025
SimpleDeepSearcher [90]	□	■	■	□	2WikiMultiHopQA	Qwen-2.5-7B-In, Qwen-2.5-32B-In, DeepSeek-Distilled-Qwen-2.5-32B, QwQ-32B	Apr-2025
Suna AI [4]	■	■	□	□	□	GPT-4o, Claude	Apr-2025
AgenticSeek [61]	□	■	□	□	□	GPT-4o, DeepSeek-R1, Claude	May-2025
Alita [75]	■	■	■	□	PathVQA	GPT-4o, Claude-Sonnet-4	May-2025
DeerFlow [19]	■	□	□	□	□	Doubao-1.5-Pro-32k, DeepSeek-R1, GPT-4o, Qwen	May-2025
PANGU DEEPDIVER [84]	■	□	□	□	C-SimpleQA, HotpotQA, ProxyQA	Pangu-7B-Reasoner	May-2025

Tool Use

- Code **Interpreter**, execute scripts during inference (Python)
- Data Analytics, transform raw retrievals into **structured** insights such as charting, table generation and statistical analysis (SQL-based queries, table-based web interfaces)
- **Multimodal** Processing and Generation, analyze heterogeneous data such as text, images, audio and video (GitHub, Notion, Google Maps, Sympy, Excel)

Tool Use

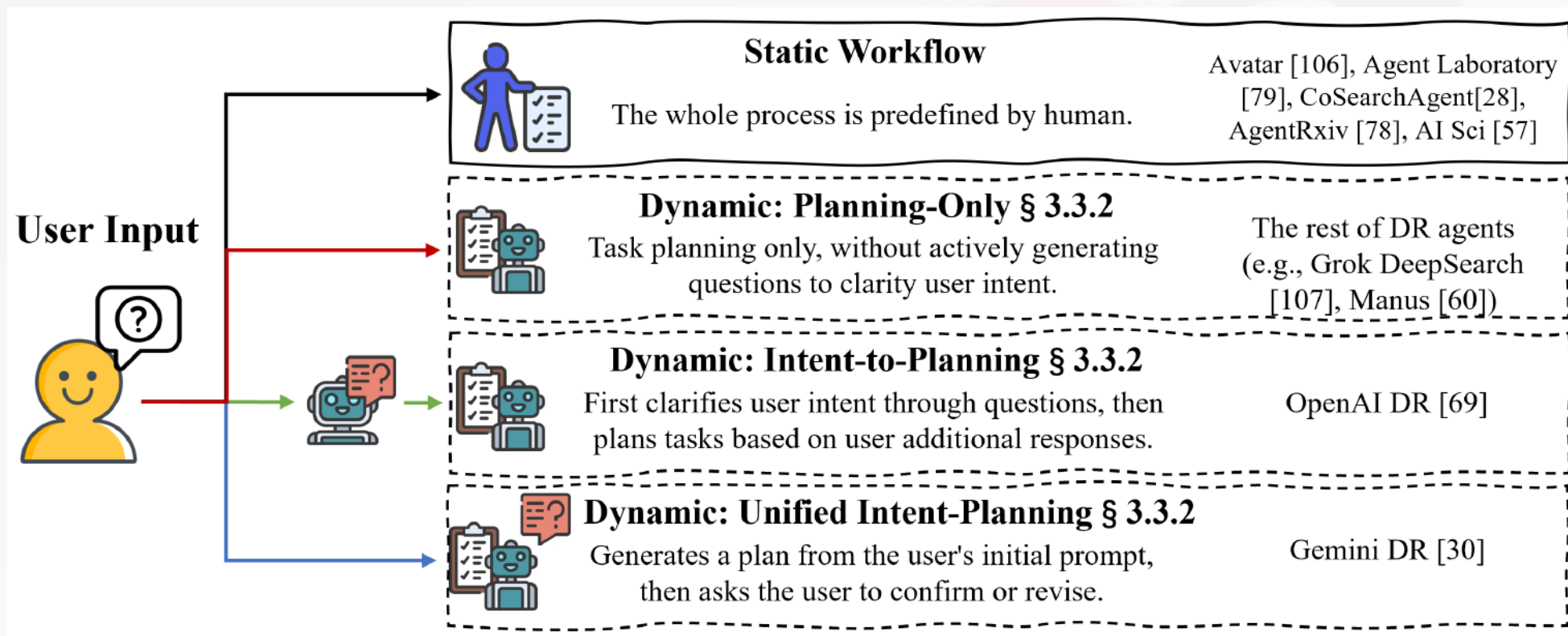
Table 2: Comparison of DR Agents with Tool Use Capabilities

■ = Involved, ■ = Non Disclosure, □ = Not present

DR Agent	Code Interpreter	Data Analytics	Multimodal	Release
CoSearchAgent [28]	□	■	□	Feb-2024
Storm [81]	■	□	□	Jul-2024
The AI Scientist [57]	■	□	□	Aug-2024
Agent Laboratory [79]	■	□	□	Jan-2025
Agentic Reasoning [105]	■	□	□	Feb-2025
AutoAgent [91]	■	□	■	Feb-2025
Genspark DR [94]	■	■	■	Feb-2025
Grok DeepSearch [107]	■	■	■	Feb-2025
OpenAI DR [69]	■	■	■	Feb-2025
Perplexity DR [72]	■	■	■	Feb-2025
Agent-R1 [71]	■	□	□	Mar-2025
AutoGLM Romination [119]	■	□	■	Mar-2025
Copilot Researcher [63]	■	■	■	Mar-2025
Manus [60]	■	■	■	Mar-2025
OpenManus [54]	■	■	□	Mar-2025
OWL [11]	■	■	■	Mar-2025
H2O.ai DR [35]	■	■	■	Mar-2025
Genspark Super Agent [94]	■	■	■	Apr-2025
WebThinker [53]	■	□	□	Apr-2025
Suna Ai [4]	■	■	□	Apr-2025
AgenticSeek [61]	■	■	□	May-2025
Alita [75]	■	■	■	May-2025
DeerFlow [19]	■	■	□	May-2025

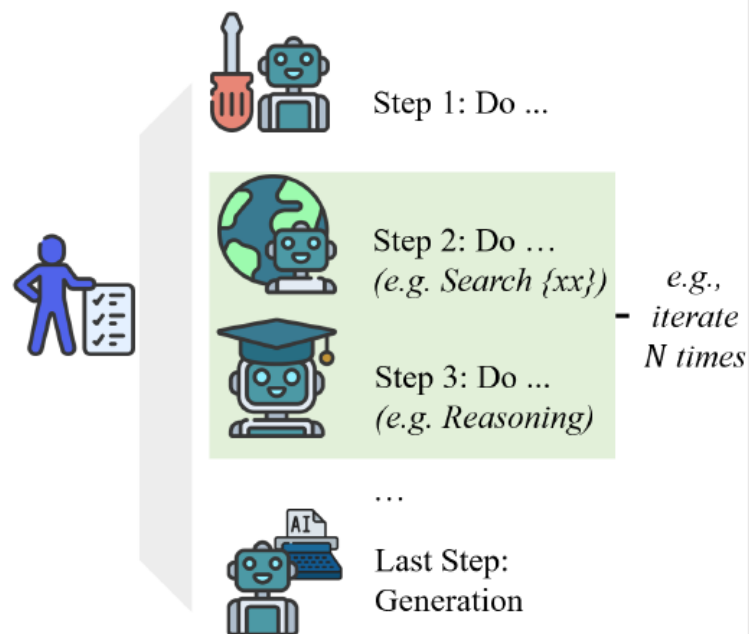
Workflow

- Static, **manually** predefined
- Dynamic, **adaptive** task planning



Workflow

Static Workflow § 3.3.1

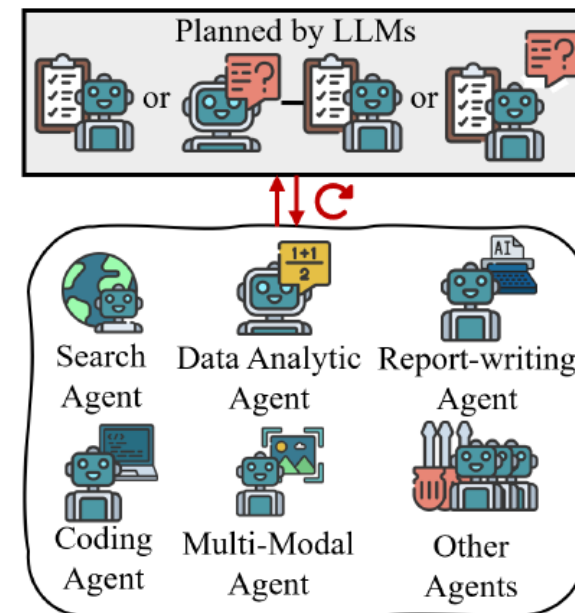


(a) **Static Workflow:** Manually predefined pipeline; fixed agent roles; simple yet limited adaptability, requiring tailored designs for each distinct task.

Dynamic Workflow



(b) **Dynamic-Single-Agent Workflow § 3.3.3:** A powerful LRM iteratively generates and updates plans, dynamically invokes tools (e.g., search, coding), and produces reports.



(c) **Dynamic-Multi-Agent Workflow § 3.3.3:** LLMs generate plans, assign tasks to agents, refine plans from real-time response, iteratively reassign tasks until produce final reports.

Dynamic Single-Agent Systems,
enable direct **end-to-end RL** (Agent-
R1, ReSearch, and Search-R1)

Dynamic Multi-Agent Systems, **hierarchy,**
coordinate, **difficult to conduct end-to-end**
RL (OpenManus, Manus, OWL, Alita)

Workflow

- Memory Mechanism for Long-Context Optimization
 - Expanding the Context **Window** Length; (Google's Gemini)
 - **Compressing** Intermediate Steps; (The AI Scientist, CycleResearcher, Search-o1)
 - Utilizing **External** Structured Storage for Temporary Results (such as Manus, OWL, Open Manus, Avatar, WebThinker, AutoAgent, Agentrxiv)

Tuning

● SFT

- Optimizing LLMs for **critical components** (search query formulation, structured report generation, and external tool utilization)
- Remain confined to **offline, static retrieval** pipelines characteristic of retrieval-augmented systems (Open-RAG, AUTO-RAG, DeepRAG, CoRAG, ATLAS)

● Reinforcement Learning

- More **adaptive, significantly strengthens** capabilities (DeepRetrieval, ReSearch, R1-Searcher, Agent-R1, WebThinker, Pangu DeepDiver)
- Reward (rule-based, final results)
- Algorithm (PPO, GRPO, Reinforce++)

Tuning

Table 3: Comparison of DR Agents with Tuning Methods

■ = Yes, ■ = Yes but details unknown, □ = Not present

DR Agent	SFT	RL	Base Model	Data	Reward Design	Release
Gemini DR [30]	■	■	Gemini-2.0-Flash	□	■	Dec-2024
Grok DeepSearch [107]	□	■	Grok3	□	■	Feb-2025
OpenAI DR [69]	□	■	GPT-o3	□	■	Feb-2025
Agent-R1 [71]	□	PPO [80], Reinforce++ [38], GRPO [82]	Qwen2.5-1.5B-Inst	HotpotQA	Rule-Outcome	Mar-2025
AutoGLM Romination [119]	■	■	GLM-Z1-Air	□	■	Mar-2025
H2O.ai DR [35]	■	■	h2ogpt-oasst1-512-12b	□	■	Mar-2025
Copilot Researcher [63]	■	■	o3-mini	□	□	Mar-2025
ReSearch [15]	□	GRPO	Qwen2.5-7B-Inst, Qwen2.5-32B-Inst	2WikiMultiHopQA	Rule-Outcome	Mar-2025
R1-Searcher [86]	■	Reinforce++, GRPO	Qwen2.5-7B-InSt, LLaMA-3.1-8B-Inst	2WikiMultiHopQA, HotpotQA	Rule-Outcome	Mar-2025
Search-R1 [47]	■	PPO, GRPO	Qwen2.5-3B/7B, LLaMA3.2-3B-Inst	NQ, HotpotQA	Rule-Outcome	Mar-2025
DeepResearcher [117]	□	GRPO	Qwen2.5-7B-Inst	NQ, HotpotQA	Rule-Outcome	Apr-2025
Gensparks Super Agent [94]	□	□	Mixture of Agents	□	□	Apr-2025
WebThinker [53]	■	Iterative Online DPO	QwQ-32B	Expert Dataset	Rule-Outcome	Apr-2025
SWIRL [27]	□	Offline-RL	Gemma-2-27B	HotPotQA	□	Apr-2025
SimpleDeepSearcher [90]	■	PPO	Qwen-2.5-7B-In, Qwen-2.5-32B-In, Deepseek-Distilled-Qwen-32B, QwQ-32B	NQ, HotpotQA, 2WikiMultiHopQA, Musique, SimpleQA, MultiHop-RAG	Process-based reward	Apr-2025
PANGU DEEPDIVER [84]	■	GRPO	Pangu-7B-Reasoner	WebPuzzle	Rule-Outcome	May-2025

Non-parametric Continual Learning

Refine their capabilities **at runtime** by optimizing external memory, workflows, and tool configurations through continuous interaction with the external environment, online adaptation with minimal data and computational overhead

- Case-based reasoning (CBR), dynamic retrieval and adaptation of structured trajectories
 - CBR operates at the trajectory level
 - emphasizes reasoning-centered memory organization
 - DS-Agent, LAM, Agent K, AgentRxiv, Kaggle Grandmaster Agent, Alita
- Prompt-based methods, encode **fixed demonstrations** or task heuristics into static input templates
- RAG, **retrieves** unstructured textual content from static corpora

Deep Research Agents: Implementations

Deep Research Agents

- OpenAI Deep Research
- Grok
- Kimi-Researcher
- Gemini Deep Research
- Perplexity Deep Research
- Microsoft 365 Copilot

OpenAI Deep Research



- A new agentic capability that conducts **multi-step research on the internet** for complex tasks
- Early version of OpenAI o3 that is **optimized for web browsing**
- Leverages reasoning to **search, interpret, and analyze** massive amounts of text, images, pivoting as needed in reaction to information it encounters
- Read files and analyze data by **writing and executing python code**

OpenAI Deep Research



- Trained on new browsing datasets created specifically for research use cases
 - Objective auto-gradable tasks with ground truth answers
 - More open-ended tasks with accompanying rubrics for grading
 - Model responses are graded against the ground truth answers or rubrics using a chain-of-thought model as a grader.
 - Through reinforcement learning
- Learned capabilities
 - Browsing capabilities (searching, clicking, scrolling, interpreting files)
 - How to use a python tool in a sandboxed setting (carrying out calculations, doing data analysis, plotting graphs)
 - How to reason through and synthesize a large number of websites to find specific pieces of information or write comprehensive reports

Safety

Risk	Mitigations
Prompt Injections	• Post-training • Preventing deep research from constructing and navigating to arbitrary URLs
Disallowed Content	• Post-training • Blocklist • Output classifiers • Output filters • Abuse monitoring and enforcement
Privacy	• Post-training • Blocklist • Abuse monitoring and enforcement
Ability to Run Code	• No access to internet • Sandboxed coding environment
Bias	• Post-training
Hallucinations	• Post-training, including increased reliance on search • Linking to inline sources for verification

Grok 3



Grok 3 Beta, blending superior reasoning with extensive pretraining knowledge

- **Significant improvements** in reasoning, mathematics, coding, world knowledge, and instruction-following tasks.
- Grok 3's reasoning capabilities, refined through **large scale reinforcement learning**, allow it to think for seconds to minutes, correcting errors, exploring alternatives, and delivering accurate answers.

Grok 3

- Grok 3 (Think) and Grok 3 mini (Think). Thinking Harder: Test-time Compute and Reasoning
 - Trained using **RL at an unprecedented scale** to **refine its chain-of-thought process**
 - Can spend anywhere **from a few seconds to several minutes** reasoning, often considering **multiple approaches**, **verifying** its own solution, and **evaluating** how to precisely meet the requirements of the problem
- Grok Agents: Combining Reasoning and Tool Use
 - Rolling out **DeepSearch**, synthesize key information, reason about conflicting facts and opinions, and distill clarity from complexity.
 - Final summary trace results in a concise and comprehensive **report**

Grok 4



Better than phd in everything (Grok 4 , Grok 4 Heavy (multi-agent), no tools/tool use)

- Colossus 200,000 GPU cluster, run **reinforcement learning** training that refines Grok's reasoning abilities **at pretraining scale**.
- New infrastructure, increase training by 6x
- A **massive data collection effort**, **significantly** expanded verifiable training data from primarily math and coding data to **many more domains**
- **Smooth** performance gains while training on **over an order of magnitude more compute** than previously
- Grok 4 Heavy, parallel test-time compute, consider **multiple hypotheses at once**

<https://www.youtube.com/watch?v=7t8NgGlGIqg>

<https://x.com/i/events/1942716886258528256>

<https://dev.to/code42cate/did-clon-musk-just-invent-agi-everything-you-need-to-know-about-grok-4-and-how-to-try-it-out-2p6m>

<https://x.ai/news/grok-4>

Grok 4

GPQA Science	LiveCodeBench (Jan - May) Competitive Coding	USAMO 2025 Olympiad Math Proofs
Grok 4 Heavy w/ Python88.4 Grok 487.5 Gemini 2.5 Pro86.4 o383.3 Claude Opus 479.6	Grok 4 Heavy w/ Python79.4 Grok 4 w/ Python79.3 Grok 479 Gemini 2.5 Pro74.2 o372	Grok 4 Heavy w/ Python61.9 Gemini Deep Think49.4 Grok 437.5 Gemini 2.5 Pro34.5 o321.7
HMMT 2025 Competitive Math	AIME'25 Competition Math	ARC-AGI-2 Abstraction and Reasoning
Grok 4 Heavy w/ Python96.7 Grok 4 w/ Python93.9 Grok 490 Gemini 2.5 Pro82.5 o377.5 Claude Opus 458.3	Grok 4 Heavy w/ Python100 Grok 4 w/ Python98.8 o3 w/ Python98.4 Grok 491.7 o388.9 Gemini 2.5 Pro88 Claude Opus 475.5	Grok 415.9 Claude Opus 48.6 o36.5 Gemini 2.5 Pro4.9

Kimi-Researcher



An **autonomous agentic and thinking model** designed to solve complex problems through multi-step planning, reasoning, and tool use.

- Built on an internal version of the Kimi k-series model
- Trained entirely through **end-to-end agentic RL**
- Leverages three main tools: a parallel, real-time internal **search** tool; a text-based **browser** tool for interactive web tasks; and a **coding** tool for automated code execution.

Formally, given the state observation s_t (for instance, s_0 includes system prompt, tool declarations, and user query), Kimi-Researcher generates $think_t$ and $action_t$. An action can either be a tool call or an indication to terminate the trajectory. The detailed behavior of Kimi-Researcher is as follows:

$$\begin{cases} (s_t) \xrightarrow{\text{Kimi-Researcher}} (think_t, action_t) \\ s_{t+1} = \text{context_manager}(s_t, think_t, \text{tool_call_result}_t) & \text{if } action_t \neq \text{finish} \\ \text{terminate} & \text{if } action_t = \text{finish} \end{cases}$$

● Challenges

- Dynamic Environments: **identical queries can yield different results** over time (distribution shifts)
- Long-Horizon Tasks: 70+ search queries, hundreds of thousands of tokens. This demands advanced **memory management** and **long-context models**.
- Data Scarcity: High-quality RL datasets for agentic QA are rare. **Automatically synthesizing training data**, large-scale learning **without manual labeling**
- Rollout Efficiency: Multi-turn reasoning and heavy tool use

Training data

- Developed a suite of challenging, **tool-centric tasks** designed to promote robust tool-use learning
 - **When** to invoke a tool
 - **How** to orchestrate tool use effectively
- Curated and synthesized **reasoning-intensive tasks**, reinforce core **cognitive** abilities and its capacity to **integrate** reasoning with tool usage.
 - Math and Code Reasoning: logical inference, algorithmic problem-solving, and sequential computation. **with toolset** beyond pure chain-of-thought.
 - Hard Search: **iteratively** search, synthesize, and reason within context constraints to derive valid answers.

Training data

- Generating
- Fully **automated pipeline**, generating and validating question-answer pairs with **minimal manual intervention**
- Robust **ground truth (GT) extraction method** to guarantee that each question is paired with a reliable answer whenever possible
- Rigorous **filtering** funnel removes ambiguous, trivial, or incorrect pairs
- **Pass@N checks** ensuring only non-trivial questions are retained

RL training: REINFORCE algorithm

- On-policy Training

- Critical to generate strict on-policy data
- Disable LLM engine mechanisms like [toolcall format enforcers](#), to ensure each trajectory is generated entirely based on the model's own probability distribution

- Negative Sample Control

- Negative samples lead to a decrease in token probabilities, which increases the [risk of entropy collapse](#) during RL training
- Discard some negative samples strategically

RL training

- Outcome rewards

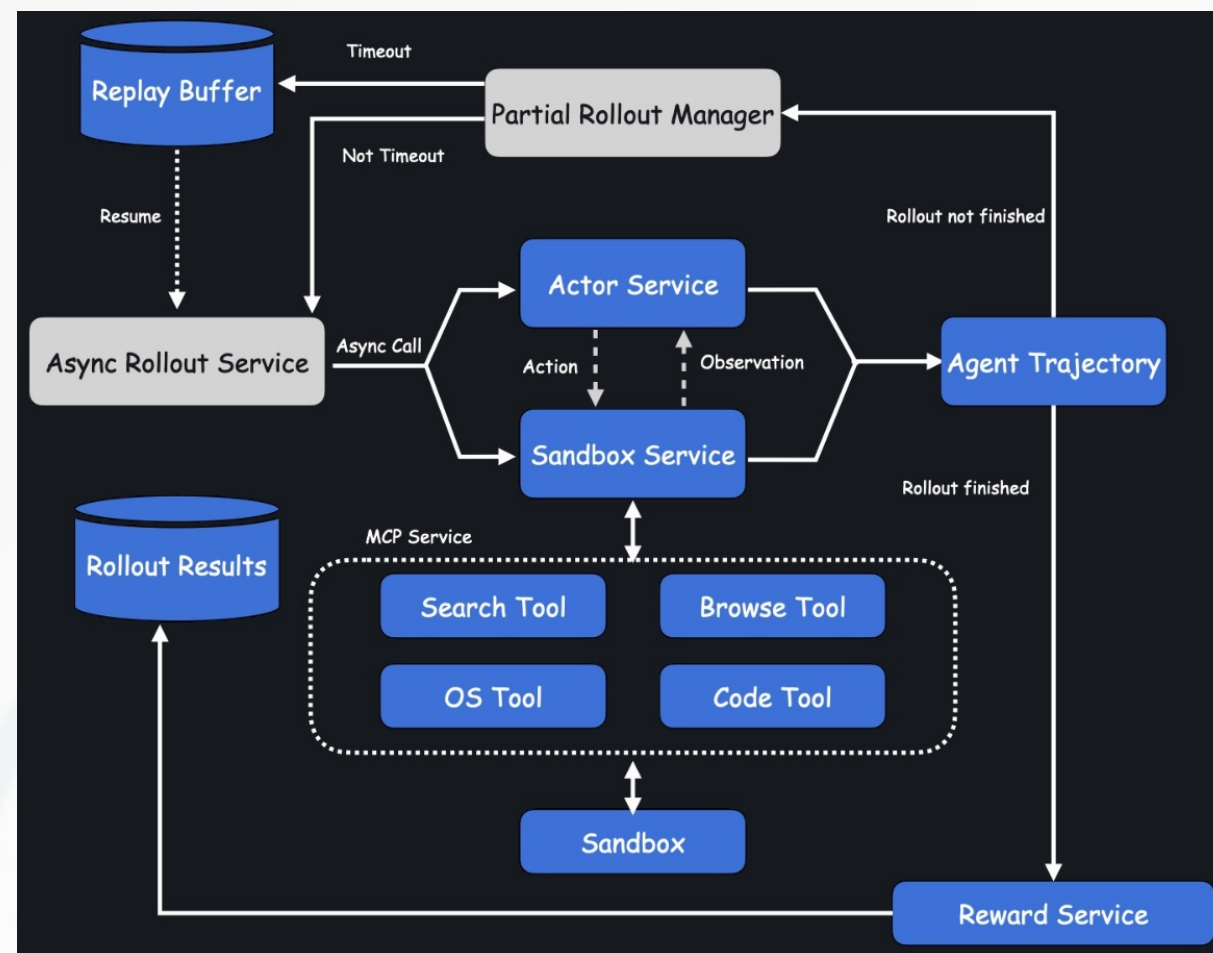
- Format Reward: penalize trajectories that include **invalid tool calls** or exceeds the maximum **limitation**
- Correctness Reward: **comparison** between the model's answer and the ground truth
- A **gamma-decay factor** is applied, $r \times \gamma^{T-i}$. Encourages the model to discover **shorter**, more efficient exploration

- Context management

- **Exceed the limitation** within **10 iterations** without memory management
- **Retain important information while discarding unnecessary documents**, extending a single rollout trajectory to **over 50 iterations**

Large-scale agent RL infra

- Fully asynchronous rollout
 - Extensible **Gym-like** interfaces
 - Orchestrates rollouts, interactions, reward calculations in **parallel**, eliminating idle time
- Turn-level Partial Rollout mechanism
 - A small fraction of tasks required **extensive turns**
 - **Save to a replay buffer**. In subsequent iterations, executed with updated model weights
- Robust sandbox environment
 - Eliminates container overhead, maintain isolation
 - Zero-downtime scheduling with Kubernetes-based hybrid cloud architecture
 - Model Context Protocol (MCP)



Gemini Deep Research



Automatically browse up to hundreds of websites on your behalf, think through its findings, and create insightful multi-page reports in minutes.

Trained Gemini to be capable of:

- **Breaking down the problem**: formulates a detailed research plan, breaking the problem. **You're in control** of the plan: can refine it to make sure it's focused on the right areas
- **Research**
 - Oversees the execution of this plan, intelligently determines which sub-tasks can be tackled **simultaneously** and which need to be done **sequentially**.
 - Can use tools like search and web browsing to fetch information, reason over it, decide next move
- **Synthesis**: synthesizes the findings into a comprehensive report, perform **multiple passes of self-critique** to enhance clarity and detail

Gemini Deep Research



Three significant technical challenges

- Multi-step planning
 - has to **ground** itself on all information gathered so far,
 - **identify** missing information and discrepancies it wants to explore
 - trading off comprehensiveness with compute and user wait time.
- Long-running inference
 - A typical Deep Research task involves many model calls, a single **failure** doesn't mean having to restart the task from the beginning.
 - **asynchronous task manager** that maintains a shared state between the planner and task models, allowing for graceful **error recovery without restarting** the entire task.
- Context management
 - Gemini's industry-leading **1 million token context** window complemented with a **RAG** setup, **"remember" everything** it has learned

Perplexity Deep Research



Performs dozens of searches, reads hundreds of sources, and reasons through the material to autonomously deliver a comprehensive report.

- Research with reasoning
 - Equipped with **search and coding** capabilities,
 - Iteratively searches, reads documents, and **reasons** about what to do next,
 - **Refine** its research plan
- Report writing - synthesizes all the research into a comprehensive **report**
- Export & Share - export the report

Microsoft 365 Copilot



Researcher - Tackle complex, multi-step research at work—delivering insights with greater quality and accuracy than previously possible.

- It combines **OpenAI's deep research** model with Microsoft 365 Copilot's advanced orchestration and deep search capabilities.

Analyst - Think like a skilled data scientist, go from raw data to insights in minutes.

- Built on **OpenAI's o3-mini reasoning model** and **optimized** to do advanced **data analysis**
- Uses **chain-of-thought reasoning** to progress through problems iteratively, taking as many steps as necessary to refine its reasoning and provide a high-quality answer
- can run **Python**

Deep Research Agents: Implementations

Opensource project
Jina AI: node-deepresearch

Jina AI: node-deepresearch

	DeepSearch	DeepResearch
解决的问题	通过迭代搜索提高信息的准确性和完整性	在文档层面上提升内容的组织性、连贯性和可读性
最终呈现	简洁的答案，附带相关网址作为参考	结构化的长篇报告，包含多个章节、图表、表格和参考文献
核心复杂度	状态机架构，具有清晰的状态转换条件；在失败的尝试中保持持久性，直至问题解决	多层次架构，同时管理微观层面（搜索）和宏观层面（文档）的关注点；采用结构化方法管理复杂的信息层级
优化重点	局部优化（最佳的下一步搜索/阅读行动）	全局优化（章节组织、术语一致性、过渡）
局限性	受限于搜索质量和推理能力	受限于 DeepSearch 的质量，以及组织复杂性和叙事连贯性方面的挑战

https://mp.weixin.qq.com/s/-pPhHDi2nz8hp5R3Lm_mww

<https://github.com/jina-ai/node-DeepResearch>

<https://mp.weixin.qq.com/s/apnorBj4TZs3-Mo23xUReQ>

https://mp.weixin.qq.com/s?__biz=MzkyODIxMjczMjA==&mid=2247501187&idx=1&sn=94fd0b8f53b39bd0adbaab86834996cd&scene=21&poc_token=HDd6dGijhZHaK1PCCad692CJL8TPSQEoqo6Xxf3o

Jina AI: node-deepresearch

DeepResearch核心在于循环推理机制。与大多数RAG系统试图一步到位地回答问题不同，Jina AI采用了一种迭代循环的方式。持续搜索信息、阅读相关来源并进行推理，直到找到答案或耗尽token预算。

```
// 主推理循环
while (tokenUsage < tokenBudget && badAttempts <= maxBadAttempts) {
  // 追踪进度
  step++; totalStep++;

  // 从 gaps 队列中获取当前问题，如果没有则使用原始问题
  const currentQuestion = gaps.length > 0 ? gaps.shift() : question;

  // 根据当前上下文和允许的操作生成提示词
  system = getPrompt(diaryContext, allQuestions, allKeywords,
    allowReflect, allowAnswer, allowRead, allowSearch, allowCoding,
    badContext, allKnowledge, unvisitedURLs);

  // 让 LLM 决定下一步行动
  const result = await LLM.generateStructuredResponse(system, messages, schema);
  thisStep = result.object;

  // 执行所选的行动（回答、反思、搜索、访问、编码）
  if (thisStep.action === 'answer') {
    // 处理回答行动...
  } else if (thisStep.action === 'reflect') {
    // 处理反思行动...
  } // ... 其他行动依此类推
}
```

●为了保证输出的稳定性和结构化，**有选择地禁用某些操作**。引导 Agent 沿着正确的方向前进，避免在原地打转。

- 当内存里没有 URL 时，禁止“visit”操作；
- 如果上次的回答被拒绝，阻止 Agent 立即重复“answer”操作。

Jina AI: node-deepresearch

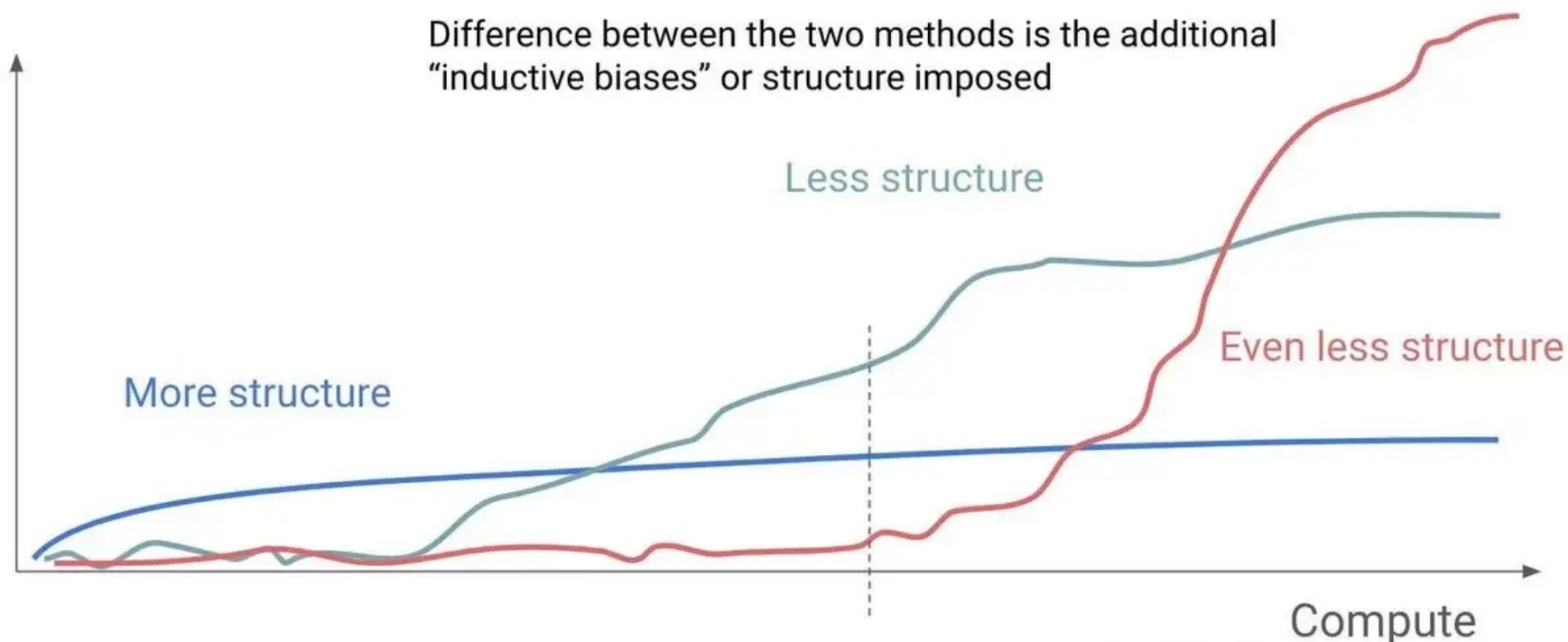
- 系统提示词
 - 使用 XML 标签定义各个部分，生成更健壮的系统提示词和生成内容
 - 直接在 JSON Schema 的 description 字段中加入字段约束，效果更好
- 知识空白问题，在回答核心问题之前，Agent 需要先补足的知识缺口，不直接尝试回答原始问题，**识别并解决子问题**
 - 带轮转机制的 FIFO（先进先出）队列
 - 新的知识空白问题会被优先推到队列头部
 - 原始问题始终位于队列尾部
 - 系统在每个步骤都从队列头部提取问题进行处理。当一个知识空白问题被解决后，获得的知识可以立即应用于所有后续问题，最终解决原始问题
- 查询重写
 - 查询重写的**重要性远超预期**
 - jina-embeddings-v3

Jina AI: node-deepresearch

- 爬取网页内容
 - Jina Reader API
 - 完整的网页内容，摘要片段，对 URL 进行规范化，限制URL访问数量
- 内存管理
 - 内存系统区分了什么算作"记忆"，什么算作"知识"
 - Agent 的记忆由上下文窗口内的三部分组成：获得的知识、访问过的网站以及失败尝试的日志
- 答案评估
 - 答案生成和评估放在不同的提示词中完成效果更好。
 - 首先确定评估标准，然后逐一进行评估。参考少量示例来进行一致性评估，比自我评估更可靠
- 预算控制
 - 尝试回答之前识别知识缺口
 - 灵活启用和禁用某些操作，引导系统运用能够加深推理的工具
 - 限制失败尝试的次数
 - 当接近预算上限时，确保无论如何都给出一个答案

Don't Teach, Incentivize

Performance



If we are here, we should choose "Less structure". But remember to undo later

5037 @Flood Sung



THANKS!

